



Docker Session III

Dockerfile

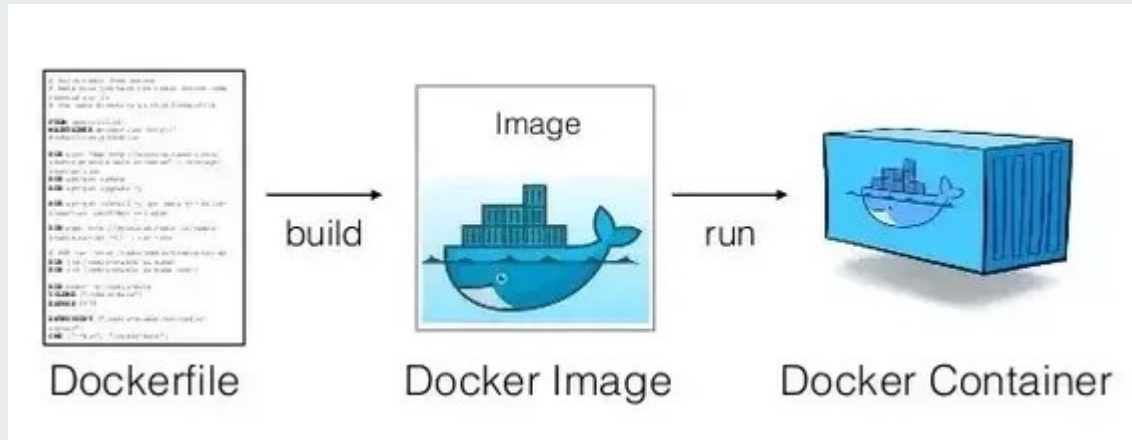
Discussion on

- Dockerfile
- Dockerfile commands
- Dockerfile demo

Dockerfile

- What is dockerfile?

Docker Workflow



Dockerfile commands

- FROM
- RUN
- ADD
- COPY
- EXPOSE
- VOLUME
- CMD
- ENTRYPOINT

FROM <base_image>

- FROM scratch
- FROM ruby:2.5.0
- FROM node:6.0

MAINTAINER <name>

- MAINTAINER Degenda SIVakoti <degendra@bajratechnologies.com>
- Deprecated

RUN <command>

- RUN apt-get install nodejs
- RUN apt-get update && apt-get install nodejs postgresql nano curl

ADD <src> <dest>

- ADD ..
- ADD ./app/
- ADD Gemfile /app/Gemfile

COPY <src> <dest>

- COPY ./client
- COPY package.json /client/package.json

ADD vs COPY

- ADD has 2 extra capabilities (either or)
 - `<src>` can be url
 - Archive files will be automatically extracted.
- Add <https://google.com/images/a.png> `/app/images/a.png`
- Add `file.tar.gz` `/app`

GUESS

- Add <https://google.com/images/a.png> /app/images/b.png
- Add <https://google.com/images/a.png> /app/images/
- Add <https://google.com/images/a.tar.gz> /app/
- Add <https://google.com/images/a.tar.gz> /app/b.tar.gz

ANSWER

- Add <https://google.com/images/a.png> /app/images/b.png
/app/images/b.png
- Add <https://google.com/images/a.png> /app/images/
/app/images/a.png
- Add <https://google.com/images/a.tar.gz> /app/
/app/a.tar.gz
- Add <https://google.com/images/a.tar.gz> /app/b.tar.gz
/app/b.tar.gz

EXPOSE <port>

- EXPOSE 80

ENV <key> <value>

- ENV RAILS_ENV development
- ENV GO_ENV production

ENV <key>=<value>

- ENV RAILS_ENV=development GO_ENV=production

Difference between <key> <value> and <key>=<value>?

VOLUME ["/data"]

- VOLUME /var/log
- VOLUME /var/log /var/db
- VOLUME ["/var/log", "/var/db"]

WORKDIR <path>

- Mkdir /demo-app
- WORKDIR /demo-app
- PWD
 - /demo-app

SHELL & EXEC

- Shell form
 - RUN <command>
 - RUN echo "Hello World!"
 - /bin/sh -c "echo 'Hello World!'"
- Exec form
 - RUN ["executable", "param1", "param2"]
 - RUN ["/bin/bash", "-c", "echo Hello World!"]

CMD

- CMD ["executable", "param1", "param2"]
- CMD ["param1", "param2"]
- CMD command param1 param2
- CMD ["/usr/bin/tail", "-f", "/dev/null"]

ENTRYPOINT

- ENTRYPOINT command param1 param2
- ENTRYPOINT echo "Hello World!"
- CMD ["Hello World!"]
- ENTRYPOINT echo
- ENTRYPOINT echo
- Docker run <image_name> "Hello World"

ENTRYPOINT Cont...

- ENTRYPOINT command param1 param2
- ENTRYPOINT echo "Hello World!"
- CMD ["Hello World!"]
- ENTRYPOINT echo //Hello World!
- ENTRYPOINT echo
- Docker run <image_name> "Hello World" //Hello World!

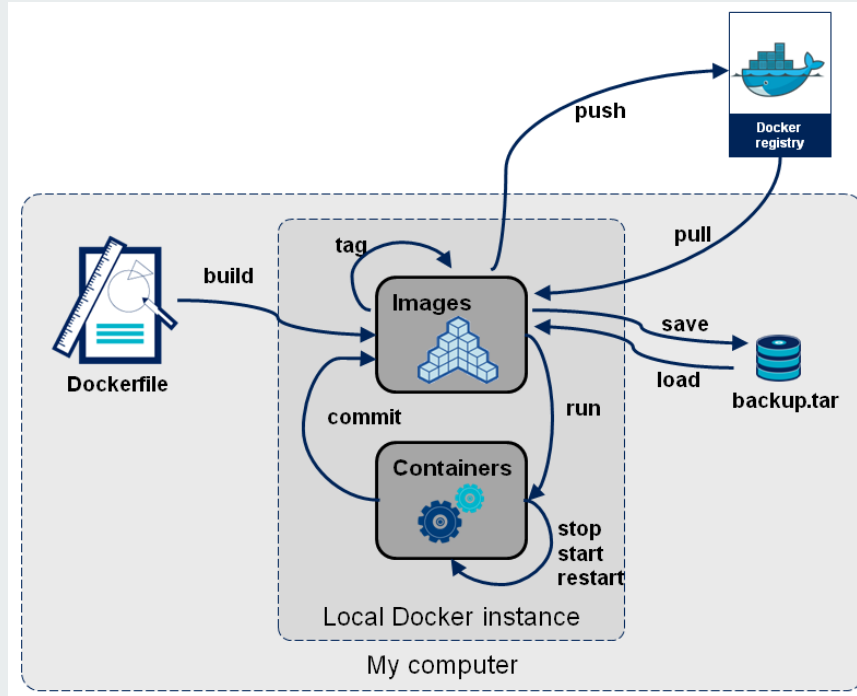
OTHERS

- LABEL
- USER
- ONBUILD

.dockerignore File

- Similar to .gitignore
- Useful when using ADD and COPY command
 - # comment
 - log/*
 - node_modules/*

Docker Workflow Detailed



Build Context

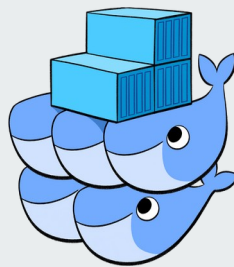
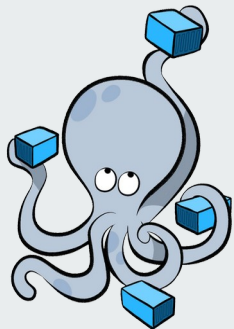
- Docker build .
- Docker build -f <dockerfile_name> <build_context>
- Docker run -d -p 8080:80 -v host/path:container/path
<image_name> <commands>

Dockerfile Demo

- Finally!!!

UPCOMING

- Docker compose
- Docker swarm





Queries???



Thank You!!!